

Functional Equations

A Brief Guide

1. Introduction

A *functional equation* is a relation that an unknown function must satisfy at every point (or pairs of points) of its domain. Solving a functional equation means finding every function compatible with the relation, not just one or two examples. The discipline is closer to algebra than to analysis: most of the work is symbolic substitution, with a thin layer of analysis (continuity, monotonicity, density) used at the end to push a relation defined on a sparse set out to all of $\mathbb{R}$.

Three skills carry almost everything in this subject. The first is choosing substitutions: setting one or both variables to a fixed value, swapping signs, replacing x by $1-x$ or $1/x$, exchanging the roles of variables. The second is the rational ladder: showing a relation on $\mathbb{N}$, then on $\mathbb{Z}$, then on $\mathbb{Q}$, then on $\mathbb{R}$, where the last step requires a regularity condition. The third is recognising structure: spotting when a relation is the additive or multiplicative Cauchy equation in disguise, when it admits an order-three Möbius cycle, or when it forces an even or odd function. The goal of this note is to make all three habitual.

2. Density of Rationals and the Approximation Lemmas

Every theorem in this note that pushes a relation from $\mathbb{Q}$ to $\mathbb{R}$ rests on the same structural fact about the real line: the rationals form a dense subset.

THEOREM. Density of $\mathbb{Q}$ in $\mathbb{R}$. For every pair $x, y \in \mathbb{R}$ with $x < y$, there exists a rational r with $x < r < y$.

The proof is standard: choose $n \in \mathbb{N}$ large enough that $n(y-x) > 1$, so the interval (nx, ny) has length greater than one and therefore contains an integer m , and set $r = m/n$.

Two consequences of density underlie everything that follows.

PROPOSITION. Approximation by rationals (R1). For every $x \in \mathbb{R}$, there exists a sequence $\{q_n\}_{n \geq 1}$ of rationals with $q_n \rightarrow x$ as $n \rightarrow \infty$.

A concrete construction is $q_n = [nx]/n$, where $[\cdot]$ denotes the greatest integer function. Writing $\{nx\} = nx - [nx]$ for the fractional part, $0 \leq \{nx\} < 1$, so

$$q_n = \frac{nx - \{nx\}}{n} = x - \frac{\{nx\}}{n}.$$

Since $|\{nx\}/n| < 1/n$, the sequence q_n converges to x . Each q_n is rational because $[nx] \in \mathbb{Z}$ and $n \in \mathbb{N}$.

PROPOSITION. Two-sided approximation (R2). For every $x \in \mathbb{R}$, there exist a strictly increasing sequence of rationals $\{p_n\}$ and a strictly decreasing sequence of rationals $\{q_n\}$ with $p_n \rightarrow x$ and $q_n \rightarrow x$.

This refinement is needed when the regularity condition is monotonicity rather than continuity. The construction uses density iteratively. First, pick a rational p_1 such that $x - 1 < p_1 < x$. Next, pick a rational p_2 such that $\max(p_1, x - 1/2) < p_2 < x$, and so on. The rational sequence q_n is constructed symmetrically, approaching x from above.

REMARK. The reason both forms are needed is the difference between continuity and monotonicity. Continuity uses any sequence converging to x ; monotonicity needs the approximating sequence to lie consistently on one side of x , so that $f(p_n) \leq f(x) \leq f(q_n)$ becomes a sandwich. Either lemma alone is not enough for both conditions.

3. Cauchy's Additive Functional Equation

The single most important functional equation is the additive Cauchy equation. It is the prototype for the entire subject and the source of the standard rational-ladder argument.

DEFINITION. Cauchy's additive equation. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies *Cauchy's additive equation* if

$$f(x + y) = f(x) + f(y) \quad \forall x, y \in \mathbb{R}. \quad (C)$$

3.1 Behaviour on the rationals

The relation (C), even without any regularity, completely determines f on the rationals once $f(1)$ is known.

THEOREM. Cauchy on $\mathbb{Q}$. If $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies (C) and $c = f(1)$, then $f(r) = cr$ for every $r \in \mathbb{Q}$.

Proof. Setting $x = y = 0$ in (C) gives $f(0) = 2f(0)$, so $f(0) = 0$. Setting $y = -x$ gives $f(0) = f(x) + f(-x)$, so $f(-x) = -f(x)$, that is, f is odd.

For $n \in \mathbb{N}$, induction on n gives $f(n) = nc$: the base $n = 1$ is the definition of c , and $f(n + 1) = f(n) + f(1) = nc + c = (n + 1)c$. Combined with $f(-x) = -f(x)$, this extends to $f(z) = zc$ for every $z \in \mathbb{Z}$.

For a positive rational $r = m/n$ with $m, n \in \mathbb{N}$, write $m = n \cdot r$. Then by the same induction on n , $f(n \cdot r) = n \cdot f(r)$, so

$$f(r) = \frac{f(m)}{n} = \frac{mc}{n} = c \cdot \frac{m}{n} = cr.$$

For negative rationals, oddness handles the sign. Thus $f(r) = cr$ for every $r \in \mathbb{Q}$. □

The example to keep in mind is that (C) alone does *not* force $f(x) = cx$ on $\mathbb{R}$. Using a Hamel basis (a basis for $\mathbb{R}$ as a vector space over $\mathbb{Q}$, whose existence requires the axiom of choice), one can construct additive functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that are not linear. Such functions are nowhere continuous and unbounded on every interval. To rule them out, one needs an additional regularity condition.

3.2 Three regularity conditions

Each of the following three conditions, in addition to (C), forces $f(x) = cx$ on all of $\mathbb{R}$.

THEOREM. Cauchy with monotonicity. If $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies (C) and f is monotone, then $f(x) = cx$ for every $x \in \mathbb{R}$, where $c = f(1)$.

Proof. Suppose f is monotonically increasing. Fix $x \in \mathbb{R}$. By the two-sided approximation lemma (R2), choose rational sequences $p_n \nearrow x$ and $q_n \searrow x$. Then $p_n \leq x \leq q_n$ for every n , so monotonicity gives

$$f(p_n) \leq f(x) \leq f(q_n).$$

Since $p_n, q_n \in \mathbb{Q}$, this is $cp_n \leq f(x) \leq cq_n$. Letting $n \rightarrow \infty$, both bounds converge to cx , so $f(x) = cx$. The monotonically decreasing case is symmetric. $\square$

THEOREM. Cauchy with continuity. If $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies (C) and f is continuous at even one point, then $f(x) = cx$ for every $x \in \mathbb{R}$, where $c = f(1)$.

Proof. Continuity at one point combined with (C) gives continuity everywhere: if f is continuous at x_0 and $h \rightarrow 0$, then

$$f(x+h) - f(x) = f(x_0+h) - f(x_0) \rightarrow 0$$

by adding and subtracting the appropriate terms via (C). So we may assume f is continuous on $\mathbb{R}$.

Fix $x \in \mathbb{R}$ and choose, by (R1), a sequence of rationals $q_n \rightarrow x$. Then $f(q_n) = cq_n$ for every n . By continuity of f ,

$$f(x) = \lim_{n \rightarrow \infty} f(q_n) = \lim_{n \rightarrow \infty} cq_n = cx.$$

$\square$

THEOREM. Cauchy with multiplicativity. If $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies (C) and also $f(xy) = f(x)f(y)$ for every $x, y \in \mathbb{R}$, then $f(x) = cx$ where $c = f(1) \in \{0, 1\}$.

Proof. The multiplicative relation gives $f(x^2) = (f(x))^2 \geq 0$. For any $r \geq 0$, write $r = (\sqrt{r})^2$, so $f(r) = (f(\sqrt{r}))^2 \geq 0$. Hence f is non-negative on $[0, \infty)$.

Now suppose $a \geq b$. Then $a - b \geq 0$, so $f(a - b) \geq 0$. By (C), $f(a - b) = f(a) - f(b)$, which gives $f(a) \geq f(b)$. Therefore f is monotonically increasing, and the previous theorem applies: $f(x) = cx$.

The constraint on c comes from substituting $x = y = 1$ into the multiplicative relation: $f(1) = (f(1))^2$, so $c = c^2$, giving $c \in \{0, 1\}$. The case $c = 0$ corresponds to $f \equiv 0$, and the case $c = 1$ to $f(x) = x$. $\square$

REMARK. Other regularity conditions also work. The most general known is that f is bounded above (or below) on a set of positive Lebesgue measure; this is a theorem of Steinhaus and is well beyond the scope of competition mathematics. For ISI and CMI, the three conditions above are the relevant ones, and continuity is by far the most common.

3.3 A non-example for limits

The function $f(x) = 1/x$ satisfies $f(xy) = f(x)f(y)$ on $\mathbb{R} \setminus \{0\}$ but has no limit at 0. The two sequences

$$x_n = \frac{1}{n} \rightarrow 0^+, \quad y_n = -\frac{1}{n} \rightarrow 0^-,$$

give $f(x_n) = n \rightarrow +\infty$ and $f(y_n) = -n \rightarrow -\infty$. Both one-sided limits exist as extended real numbers but disagree, so $\lim_{x \rightarrow 0} 1/x$ does not exist. The lesson for the present subject is that a multiplicative or additive relation on a punctured domain says nothing about behaviour at the deleted point, and in particular does not force continuity there.

4. Multiplicative Variants of Cauchy

The additive Cauchy equation has three companions, obtained by applying $\ln$ or $\exp$ to one or both sides. Each is reducible to (C), so each inherits its solution under continuity. Throughout this section assume f is continuous and not identically zero.

THEOREM. Variant A: exponential Cauchy. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous, $f(x+y) = f(x)f(y)$ and $f \not\equiv 0$, then $f(x) = e^{ax}$ for some constant $a \in \mathbb{R}$.

Derivation. The relation $f(x+y) = f(x)f(y)$ with $x = y$ gives $f(2x) = (f(x))^2 \geq 0$, so $f \geq 0$ everywhere. Combined with $f(0) = 1$ (which follows from $f(0) = (f(0))^2$ together with $f \not\equiv 0$ ruling out $f(0) = 0$), continuity, and the chain of arguments below, we get $f > 0$.

Take logarithms: $\ln f(x+y) = \ln f(x) + \ln f(y)$. Set $g(x) = \ln f(x)$, which is continuous because f is continuous and positive. Then g satisfies (C) with continuity, so $g(x) = ax$ for some a , and $f(x) = e^{ax}$. $\square$

THEOREM. Variant B: logarithmic Cauchy. If $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ is continuous and $f(xy) = f(x) + f(y)$, then $f(x) = c \ln x$ for some constant $c \in \mathbb{R}$.

Derivation. The substitution $x \mapsto e^u$, $y \mapsto e^v$ converts the relation to $f(e^{u+v}) = f(e^u) + f(e^v)$. Set $g(u) = f(e^u)$, which is continuous because f and $\exp$ are. Then g satisfies (C) with continuity, so $g(u) = cu$, that is $f(e^u) = cu$. Replacing u by $\ln x$ gives $f(x) = c \ln x$. The constant is $c = f(e)$. $\square$

THEOREM. Variant C: power Cauchy on $\mathbb{R}^+$. If $f: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is continuous and $f(xy) = f(x)f(y)$, then $f(x) = x^c$ for some constant $c \in \mathbb{R}$.

Derivation. Take logarithms: $\ln f(xy) = \ln f(x) + \ln f(y)$. Set $g = \ln f$. Then g is continuous and $g(xy) = g(x) + g(y)$, so by Variant B, $g(x) = c \ln x$, giving $f(x) = e^{c \ln x} = x^c$. $\square$

4.1 Sign care on $\mathbb{R} \setminus \{0\}$

Variant C extended to all nonzero reals splits into two solution families. Setting $x = y = -t$ in $f(xy) = f(x)f(y)$ for $t > 0$ gives $f(t^2) = (f(-t))^2$, while setting $x = y = t$ gives $f(t^2) = (f(t))^2$. Hence $(f(-t))^2 = (f(t))^2$, so $f(-t) = \pm f(t)$. Continuity, applied carefully on connected components of $\mathbb{R} \setminus \{0\}$, picks out one sign on the negative axis, giving the two continuous families

$$f(x) = |x|^c \quad \text{or} \quad f(x) = \operatorname{sgn}(x) |x|^c.$$

The first is even, the second is odd. Both reduce to x^c on $\mathbb{R}^+$.

5. Building Up From Scratch: $f(x+y) = f(x)f(y)$

To illustrate the method without quoting Cauchy as a black box, we derive the structure of solutions to

$$f(x+y) = f(x)f(y), \quad f(0) = 1,$$

step by step. The argument is a model for many competition problems where induction and rational approximation are needed but the substitutions are mechanical. Throughout, write $c = f(1)$.

5.1 Step (i): positivity

PROPOSITION. $f(x) > 0$ for every $x \in \mathbb{R}$.

Proof. Setting $x = y = t/2$ in the relation gives

$$f(t) = f(t/2) f(t/2) = (f(t/2))^2 \geq 0.$$

Suppose $f(x_0) = 0$ for some $x_0 \in \mathbb{R}$. Then

$$1 = f(0) = f(x_0 + (-x_0)) = f(x_0) f(-x_0) = 0,$$

a contradiction. Hence f never vanishes, and combined with the squared expression, $f(x) > 0$ for every x . $\square$

5.2 Step (ii): values on $\mathbb{N}$

PROPOSITION. $f(n) = c^n$ for every $n \in \mathbb{N}$.

Proof. Induction on n . The base case $n = 1$ is the definition of c . Assuming $f(k) = c^k$, the relation gives

$$f(k+1) = f(k+1) = f(k) f(1) = c^k \cdot c = c^{k+1}.$$

$\square$

5.3 Step (iii): values on $\mathbb{Z}$

PROPOSITION. $f(z) = c^z$ for every $z \in \mathbb{Z}$.

Proof. Setting $y = -x$ in the relation gives $f(0) = f(x)f(-x)$, so $f(-x) = 1/f(x)$. (This division is legitimate because Step (i) shows f is positive, hence nonzero.) For a negative integer $z = -n$ with $n \in \mathbb{N}$,

$$f(z) = f(-n) = \frac{1}{f(n)} = \frac{1}{c^n} = c^{-n} = c^z.$$

Combined with Step (ii) and $f(0) = 1 = c^0$, the formula holds on all of $\mathbb{Z}$. $\square$

5.4 Step (iv): values on $\mathbb{Q}$

PROPOSITION. $f(r) = c^r$ for every $r \in \mathbb{Q}$.

Proof. A sub-induction first establishes $f(nx) = (f(x))^n$ for every $n \in \mathbb{N}$ and $x \in \mathbb{R}$. Base $n = 1$ is trivial. Assuming $f(kx) = (f(x))^k$,

$$f((k+1)x) = f(kx + x) = f(kx) f(x) = (f(x))^k \cdot f(x) = (f(x))^{k+1}.$$

Now let $r = m/n$ with $m \in \mathbb{Z}$ and $n \in \mathbb{N}$. Apply the sub-induction with $x = r$:

$$f(nr) = (f(r))^n.$$

But $nr = m \in \mathbb{Z}$, so by Step (iii), $f(nr) = c^m$. Hence $(f(r))^n = c^m$. Since $f(r) > 0$ by Step (i), take the positive n -th root: $f(r) = c^{m/n} = c^r$. $\square$

5.5 Step (v): values on $\mathbb{R}$, assuming continuity

PROPOSITION. If f is continuous, then $f(x) = c^x$ for every $x \in \mathbb{R}$.

Proof. Let $x \in \mathbb{R}$. By (R1), choose rationals $q_n \rightarrow x$. From Step (iv), $f(q_n) = c^{q_n}$ for every n . Both f and the function $t \mapsto c^t$ are continuous, so

$$f(x) = \lim_{n \rightarrow \infty} f(q_n) = \lim_{n \rightarrow \infty} c^{q_n} = c^{\lim q_n} = c^x.$$

Setting $a = \ln c$ (valid because $c > 0$ by Step (i)), this can equally be written $f(x) = e^{ax}$. $\square$

6. The Substitution Toolkit

Most competition functional equations are solved by clever substitution. The table below collects the standard moves, organised by the structural goal they achieve. The “what to look for” column gives the visual cue that suggests the substitution.

Goal	Substitution	What it gives
Find $f(0)$	$x = y = 0$	A relation in $f(0)$ alone, often a quadratic that pins it down.
Find $f(1)$	$x = y = 1$	A relation in $f(1)$ alone.
Reduce one variable	$y = 0$ or $y = 1$	Often gives $f(x)$ in terms of f at fixed points.
Test parity	$y = -x$	Relates $f(x)$ and $f(-x)$; often gives $f(0)$ or shows f is even or odd.
Self-relation	$x = y$	Yields $f(2x)$ in terms of $f(x)$, basis for induction.
Symmetry exchange	swap $x \leftrightarrow y$	Useful when the right-hand side is asymmetric in x, y .
Convert additive to multiplicative	$x \mapsto e^x, y \mapsto e^y$	Converts $f(xy) = \dots$ involving products into a $f(x+y) = \dots$ form.
Convert multiplicative to additive	take $\ln$ of both sides; set $g = \ln f$	Reduces multiplicative Cauchy to additive Cauchy, provided $f > 0$.
Reciprocal substitution	$x \mapsto 1/x$	Pairs $f(x)$ with $f(1/x)$; typically gives a 2×2 linear system.
Sign substitution	$x \mapsto -x$	Pair the original with the new equation; solve as a 2×2 linear system.
Shift substitution	$x \mapsto 1 - x$ or $x \mapsto 1 + x$	Used when the equation involves $f(1 \pm x)$; gives a second equation in $f(x)$ and $f(1 \pm x)$.
Möbius three-cycle	$x \mapsto \frac{1}{1-x}$, then $x \mapsto 1 - \frac{1}{x}$	Composing this substitution thrice returns x . Three equations, three unknowns.

Goal	Substitution	What it gives
Decoupling $f(xy) = xf(y) + yf(x)$	divide by xy , set $g(x) = f(x)/x$	Reduces to additive Cauchy in g .
Build a recurrence	set $y = x, y = 2x$, etc.	Gives $f(nx)$ in terms of $f(x)$; combined with $f(0)$ and $f(1)$, builds the rational ladder.
Test trivial solutions	$f \equiv 0, f \equiv 1, f(x) = x, f(x) = x^c, f(x) = e^{cx}$	Eliminates trivial families before doing harder work; each candidate must satisfy the original.
Composition	$x \mapsto f(x)$	Useful when the equation involves $f(f(x))$ or $f(1 + f(x))$; iterate the relation.
Fixed point	solve $f(a) = a$	Sometimes the only candidate; check whether the equation forces a fixed point.

7. Worked Problems

The problems below are arranged roughly in increasing difficulty. They cover the main techniques: shift substitution, Möbius cycles, sign substitution, the rational ladder, conversion between additive and multiplicative form, and parity arguments.

PROBLEM 1. Does there exist a function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(1 + f(x)) = 1 - x \quad \text{and} \quad f(f(x)) = x?$$

Solution. Suppose such an f exists. Replace x by $f(x)$ in the first relation:

$$f(1 + f(f(x))) = 1 - f(x).$$

Since $f(f(x)) = x$, this becomes $f(1 + x) = 1 - f(x)$, that is,

$$f(1 + x) + f(x) = 1. \quad (*)$$

Setting $x = 0$ in $(*)$ gives $f(1) + f(0) = 1$. Setting $x = 1$ gives $f(2) + f(1) = 1$. Subtracting these two equations gives $f(2) = f(0)$. Now apply f to both sides:

$$f(f(2)) = f(f(0)), \quad \text{which by the second relation gives} \quad 2 = 0,$$

a contradiction. Hence no such f exists. $\square$

PROBLEM 2. Find all $f: \mathbb{R} \setminus \{0, 1\} \rightarrow \mathbb{R}$ satisfying

$$f(x) + f\left(\frac{1}{1-x}\right) = x.$$

Solution. Let $\sigma(x) = 1/(1-x)$. Compute the iterates: $\sigma^2(x) = \sigma(1/(1-x)) = 1/(1-1/(1-x)) = (1-x)/(-x) = 1-1/x$, and $\sigma^3(x) = \sigma(1-1/x) = 1/(1-(1-1/x)) = x$. So σ has order three, cycling through $x, 1/(1-x), 1-1/x$.

The original equation reads $f(x) + f(\sigma(x)) = x$. Substituting $\sigma(x)$ for x :

$$f(\sigma(x)) + f(\sigma^2(x)) = \sigma(x) = \frac{1}{1-x}. \quad (1)$$

Substituting $\sigma^2(x)$ for x :

$$f(\sigma^2(x)) + f(x) = \sigma^2(x) = 1 - \frac{1}{x}. \quad (2)$$

Adding the original equation and (2), then subtracting (1), eliminates $f(\sigma(x))$ and $f(\sigma^2(x))$:

$$2f(x) = x + \left(1 - \frac{1}{x}\right) - \frac{1}{1-x} = \frac{x \cdot x \cdot (1-x) + (1-x)(x-1) \cdot 1 - x \cdot 1}{x(1-x)}.$$

After algebraic simplification, the numerator is $x^3 - x + 1$ and the denominator is $2x(x-1)$, giving

$$f(x) = \frac{x^3 - x + 1}{2x(x-1)}.$$

Verification by direct substitution into the original equation confirms this is the unique solution. $\square$

PROBLEM 3. Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$f(xy) = xf(x) + yf(y) \quad \forall x, y \in \mathbb{R}.$$

Prove that $f \equiv 0$.

Solution. Set $x = y = 1$: $f(1) = f(1) + f(1)$, so $f(1) = 0$. Now set $y = 1$ in the original:

$$f(x) = xf(x) + 1 \cdot f(1) = xf(x),$$

which rearranges to $(x-1)f(x) = 0$. For every $x \neq 1$, this forces $f(x) = 0$, and we already have $f(1) = 0$. Hence $f(x) = 0$ for every $x \in \mathbb{R}$. $\square$

PROBLEM 4. Determine whether there exists a one-one function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2) - (f(x))^2 \geq \frac{1}{4} \quad \forall x \in \mathbb{R}.$$

Solution. Suppose such an f exists. Setting $x = 0$ in the inequality gives $f(0) - (f(0))^2 \geq 1/4$, which rearranges to

$$(f(0) - \frac{1}{2})^2 \leq 0.$$

The only way a square can be at most zero is to equal zero, so $f(0) = 1/2$. Setting $x = 1$ gives, by the same algebra, $f(1) = 1/2$. But then $f(0) = f(1)$, contradicting injectivity. Hence no such one-one f exists. $\square$

PROBLEM 5. Let $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$ be continuous and satisfy

$$f(x+y) = \frac{f(x)f(y)}{f(x) + f(y)}$$

whenever $f(x) + f(y) \neq 0$. Find all such f .

Solution. Take reciprocals on both sides:

$$\frac{1}{f(x+y)} = \frac{f(x) + f(y)}{f(x)f(y)} = \frac{1}{f(x)} + \frac{1}{f(y)}.$$

Set $g(x) = 1/f(x)$. Then g is continuous (composition of f with the continuous function $1/t$ on $\mathbb{R} \setminus \{0\}$) and satisfies (C): $g(x+y) = g(x) + g(y)$. By the continuous Cauchy theorem, $g(x) = cx$ for some $c \in \mathbb{R}$, so

$$f(x) = \frac{1}{cx}, \quad c = \frac{1}{f(1)}.$$

□

PROBLEM 6. Solve for $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$:

$$xf(x) + 2xf(-x) = -1.$$

Solution. Replace x by $-x$:

$$-xf(-x) + 2(-x)f(x) = -1, \quad \text{that is,} \quad xf(-x) + 2xf(x) = 1.$$

Now we have the system

$$xf(x) + 2xf(-x) = -1, \quad 2xf(x) + xf(-x) = 1.$$

Adding gives $3xf(x) + 3xf(-x) = 0$, so $f(-x) = -f(x)$, that is, f is odd. Substitute $f(-x) = -f(x)$ into the original equation:

$$xf(x) - 2xf(x) = -1, \quad \text{so} \quad -xf(x) = -1, \quad \text{giving} \quad f(x) = \frac{1}{x}.$$

Verification: $x \cdot 1/x + 2x \cdot (-1/x) = 1 - 2 = -1$.

□

PROBLEM 7. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous with

$$f(x+y)f(x-y) = (f(x)f(y))^2, \quad f(0) = 1.$$

Find f .

Solution. Setting $x = 0$ in the relation gives

$$f(y)f(-y) = (f(0)f(y))^2 = (f(y))^2,$$

which rearranges (provided $f(y) \neq 0$) to $f(-y) = f(y)$. So f is even, modulo the question of whether f has any zeros. We will show shortly that $f > 0$ everywhere, justifying the division.

Setting $x = y$ in the relation gives $f(2x)f(0) = (f(x)f(x))^2$, that is,

$$f(2x) = (f(x))^4.$$

This is positive (a fourth power), so $f(2x) \geq 0$ for every x . Now we must ensure $f(x) \neq 0$. Suppose $f(x_0) = 0$ for some $x_0 \in \mathbb{R}$. Then $f(x_0) = f(2 \cdot \frac{x_0}{2}) = (f(\frac{x_0}{2}))^4 = 0$, implying $f(\frac{x_0}{2}) = 0$. Iterating this logic yields $f(\frac{x_0}{2^n}) = 0$ for all $n \in \mathbb{N}$. By continuity at $x = 0$, taking the limit as $n \rightarrow \infty$ gives $f(0) = 0$, which contradicts our initial condition $f(0) = 1$. Therefore, $f(x) \neq 0$ for all $x \in \mathbb{R}$. So, $f(2x) > 0$ for every x . Since every real number is of the form $2x$ for some x , $f > 0$ on all of $\mathbb{R}$.

We now show by induction on n that $f(nx) = (f(x))^{n^2}$ for every $n \in \mathbb{N}$ and $x \in \mathbb{R}$. Base $n = 1$ is trivial. Assuming the formula holds for $n = 1, 2, \dots, k$, the relation with y replaced by kx and x replaced by kx gives... a cleaner approach is to use the identity $f((k+1)x)f((k-1)x) = (f(kx)f(x))^2$, which follows from setting $x \rightarrow kx$, $y \rightarrow x$ in the original. Then

$$f((k+1)x) = \frac{(f(kx))^2(f(x))^2}{f((k-1)x)} = \frac{(f(x))^{2k^2} \cdot (f(x))^2}{(f(x))^{(k-1)^2}} = (f(x))^{2k^2+2-(k-1)^2} = (f(x))^{(k+1)^2},$$

completing the induction.

Setting $x = 1$ gives $f(n) = (f(1))^{n^2}$. Extending to $\mathbb{Z}$ via $f(-n) = f(n)$ (evenness) gives $f(z) = (f(1))^{z^2}$ for $z \in \mathbb{Z}$. For $r = m/n \in \mathbb{Q}$, the same induction applied with $x = r$ gives $f(nr) = (f(r))^{n^2}$, that is, $(f(1))^{m^2} = (f(r))^{n^2}$, so $f(r) = (f(1))^{(m/n)^2} = (f(1))^{r^2}$ (taking the positive root, since $f > 0$).

By continuity and (R1), the formula extends to all of $\mathbb{R}$:

$$f(x) = (f(1))^{x^2} = e^{ax^2}, \quad a = \ln f(1).$$

□

PROBLEM 8. Find all continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(x+y) + f(x-y) = 2f(x)f(y) \quad \forall x, y \in \mathbb{R}.$$

(This is d'Alembert's functional equation.)

Solution. Setting $x = y = 0$ gives $2f(0) = 2(f(0))^2$, so $f(0)(f(0) - 1) = 0$. Either $f(0) = 0$ or $f(0) = 1$.

If $f(0) = 0$, set $y = 0$: $f(x) + f(x) = 2f(x)f(0) = 0$, so $f \equiv 0$.

Assume $f(0) = 1$. Setting $x = 0$ gives $f(y) + f(-y) = 2f(0)f(y) = 2f(y)$, so $f(-y) = f(y)$, that is, f is even.

To find the nontrivial solutions, consider three cases according to the value of $f(1)$.

Case 1: $|f(1)| < 1$. Write $f(1) = \cos \alpha$ for some $\alpha \in (0, \pi)$. Setting $x = y$ in the equation gives $f(2x) + 1 = 2(f(x))^2$, that is, $f(2x) = 2(f(x))^2 - 1$. We claim $f(x) = \cos(\alpha x)$ for every $x \in \mathbb{R}$.

By induction, the relation gives $f(n) = \cos(n\alpha)$ for $n \in \mathbb{N}$ (using the Chebyshev recurrence $\cos((n+1)\alpha) = 2\cos \alpha \cos(n\alpha) - \cos((n-1)\alpha)$, which matches the recurrence $f(n+1) = 2f(1)f(n) - f(n-1)$ obtained by setting $x = n$, $y = 1$ and using $f(n-1) = f(-(n-1))$ via evenness). Extending to $\mathbb{Q}$ by the rational ladder and to $\mathbb{R}$ by continuity gives $f(x) = \cos(\alpha x)$.

Case 2: $|f(1)| > 1$. Write $f(1) = \cosh \beta$ for some $\beta > 0$. The same recurrence, with hyperbolic cosines, gives $f(x) = \cosh(\beta x)$.

Case 3: $f(1) = \pm 1$. If $f(1) = 1$, the limiting case of Case 1 (or 2) gives $f \equiv 1$. If $f(1) = -1$, one can show $f(x) = \cos(\pi x)$, the boundary of Case 1 with $\alpha = \pi$.

The complete list of continuous solutions is

$$f \equiv 0, \quad f(x) = \cos(\alpha x), \quad f(x) = \cosh(\beta x),$$

for parameters $\alpha, \beta \in \mathbb{R}$.

□

PROBLEM 9. Find all continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(x+y) = f(x) + f(y) + xy \quad \forall x, y \in \mathbb{R}.$$

Solution. The presence of the xy term suggests that the leading behaviour of f is quadratic. Let $g(x) = f(x) - x^2/2$. Then

$$g(x+y) = f(x+y) - \frac{(x+y)^2}{2} = f(x) + f(y) + xy - \frac{x^2 + 2xy + y^2}{2} = f(x) - \frac{x^2}{2} + f(y) - \frac{y^2}{2} = g(x) + g(y).$$

So g satisfies (C) and is continuous (because f is and $x^2/2$ is). By the continuous Cauchy theorem, $g(x) = cx$ for some $c \in \mathbb{R}$, hence

$$f(x) = \frac{x^2}{2} + cx.$$

The constant is $c = f(1) - 1/2$. □

PROBLEM 10. Find all continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(x+y) + f(x-y) = 2f(x) + 2f(y) \quad \forall x, y \in \mathbb{R}.$$

(This is the parallelogram law.)

Solution. Setting $x = y = 0$ gives $2f(0) = 4f(0)$, so $f(0) = 0$. Setting $x = 0$ gives $f(y) + f(-y) = 2f(y)$, so $f(-y) = f(y)$ and f is even.

Setting $y = x$ gives $f(2x) + f(0) = 2f(x) + 2f(x)$, that is, $f(2x) = 4f(x)$. By induction, $f(nx) = n^2 f(x)$ for every $n \in \mathbb{N}$, with the inductive step using

$$f((n+1)x) + f((n-1)x) = 2f(nx) + 2f(x),$$

giving $f((n+1)x) = 2n^2 f(x) + 2f(x) - (n-1)^2 f(x) = (n+1)^2 f(x)$.

Setting $x = 1$ gives $f(n) = n^2 f(1)$. Extending by evenness to $\mathbb{Z}$ and by the ladder to $\mathbb{Q}$ gives $f(r) = r^2 f(1)$ for $r \in \mathbb{Q}$. By continuity and (R1),

$$f(x) = cx^2, \quad c = f(1).$$

□

PROBLEM 11. Find all $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ continuous and satisfying

$$f(xy) = xf(y) + yf(x) \quad \forall x, y \in \mathbb{R}^+.$$

Solution. The right-hand side is symmetric in x, y , so the equation has built-in symmetry; nothing new comes from swapping. The form of the right-hand side suggests dividing by xy :

$$\frac{f(xy)}{xy} = \frac{f(y)}{y} + \frac{f(x)}{x}.$$

Set $g(x) = f(x)/x$, defined and continuous on $\mathbb{R}^+$. Then $g(xy) = g(x) + g(y)$, which is the logarithmic Cauchy equation. By Variant B,

$$g(x) = c \ln x, \quad c = g(e) = f(e)/e,$$

hence

$$f(x) = cx \ln x.$$

Verification: $c \cdot xy \ln(xy) = c \cdot xy(\ln x + \ln y) = x \cdot (cy \ln y) + y \cdot (cx \ln x) = xf(y) + yf(x)$. □

PROBLEM 12. Find all $f: \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ satisfying

$$f\left(x + \frac{y}{x}\right) = f(x) + \frac{f(y)}{f(x)} + 2y \quad \forall x, y \in \mathbb{Q}^+.$$

(This problem is from IMO 2010.)

Solution. The form of the equation, with $f(y)/f(x) + 2y$ on the right, is suggestive: the candidate solution $f(x) = x^2$ gives

$$f\left(x + \frac{y}{x}\right) = \left(x + \frac{y}{x}\right)^2 = x^2 + 2y + \frac{y^2}{x^2} = f(x) + 2y + \frac{f(y)}{f(x)},$$

which matches. So $f(x) = x^2$ is a solution. We show it is the only one.

Setting $y = x$ gives $f(x+1) = f(x) + 1 + 2x$, that is, $f(x+1) - f(x) = 2x + 1$. Iterating from $x = 1$:

$$f(n) = f(1) + \sum_{k=1}^{n-1} (2k+1) = f(1) + n^2 - 1.$$

We now show $f(1) = 1$. Setting $x = y = 1$ in the original gives

$$f(2) = f(1) + \frac{f(1)}{f(1)} + 2 = f(1) + 3,$$

which agrees with the iteration: $f(2) = f(1) + 4 - 1 = f(1) + 3$. To pin down $f(1)$, set $x = 1$ in the original:

$$f(1+y) = f(1) + \frac{f(y)}{f(1)} + 2y.$$

On the other hand, the iteration with y replacing x gave $f(y+1) = f(y) + 2y + 1$. Equating:

$$f(y) + 2y + 1 = f(1) + \frac{f(y)}{f(1)} + 2y, \quad \text{so} \quad f(y) - \frac{f(y)}{f(1)} = f(1) - 1,$$

that is, $f(y)(1 - 1/f(1)) = f(1) - 1$. If $f(1) \neq 1$, divide to get $f(y)(f(1) - 1)/f(1) = f(1) - 1$, so $f(y) = f(1)$ for every y , contradicting $f(2) = f(1) + 3$. Hence $f(1) = 1$.

Combined with the iteration, $f(n) = n^2$ for every $n \in \mathbb{N}$. Extending to $\mathbb{Q}^+$ requires more work; the standard route is to set $x = p/q$ with y chosen so that $x + y/x$ is an integer, then solve. The conclusion is $f(x) = x^2$ for every $x \in \mathbb{Q}^+$. $\square$

8. Exercises

The following page problems consolidate the techniques of this note. The first few are routine; the later ones reward careful substitution and a clear sense of which regularity condition is in force.

1. Find all continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x+y) = f(x) + f(y) + 3xy(x+y)$.
2. Find all $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f(xy + x + y) = f(xy) + f(x) + f(y)$ for all $x, y \in \mathbb{R}$.
3. Find all $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f(x^2 - y^2) = (x - y)(f(x) + f(y))$ for all $x, y \in \mathbb{R}$.
4. Find all continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x+y) = f(x)e^y + f(y)e^x$ for all $x, y \in \mathbb{R}$.
5. Determine all $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x)f(y) - f(xy) = x + y$ for all $x, y \in \mathbb{R}$.
6. Find all $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f(x+y) + f(x)f(y) = f(xy) + f(x) + f(y)$ for all $x, y \in \mathbb{R}$.
7. Find all $f: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$ satisfying

$$f(x) + f\left(\frac{x-1}{x}\right) = 1 + x \quad \text{for all } x \in \mathbb{R} \setminus \{0, 1\}.$$

8. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be such that $f(x+y) = f(x) + f(y)$ and $f(x)f(1/x) = 1$ for every $x \neq 0$. Show that $f(x) = x$ for every $x \in \mathbb{R}$. (Continuity is not assumed.)

9. Find all continuous $f: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $f(xf(y)) = yf(x)$ for all $x, y \in \mathbb{R}^+$.
10. Find all $f: \mathbb{N} \rightarrow \mathbb{N}$ satisfying $f(f(n)) + f(n) = 2n + 3$ for every $n \in \mathbb{N}$ and $f(1) = 2$. (*This is IMO 1977, Problem 6.*)

9. Closing Remarks

A few habits will save time and prevent errors on competition functional equations.

The first is to check candidate solutions. Substitutions are not equivalence transformations: each one only uses a special case of the relation, and the chain of substitutions can introduce extraneous solutions or lose constraints. After deriving a candidate f , substitute it back into the original equation and verify the relation holds for all x, y in the domain. If the verification fails, either the candidate is wrong or the original equation has no solution.

The second is to be explicit about the regularity condition. The same equation can have wildly different solution sets depending on whether one assumes continuity, monotonicity, multiplicativity, or no regularity at all. The Cauchy equation is the canonical example: without regularity, the Hamel-basis pathologies are present, and the conclusion $f(x) = cx$ does not hold. State the regularity condition clearly at the start of the solution and use it explicitly when bridging from $\mathbb{Q}$ to $\mathbb{R}$.

The third is to recognise structure. A relation involving $f(x + y)$ on the left is almost always reducible to additive Cauchy or one of its variants; a relation involving $f(xy)$ to multiplicative Cauchy. Möbius substitutions of order three or two often appear in problems with $f(1/(1-x))$ or $f((x-1)/x)$. The fixed-point method is useful when the equation involves $f(f(x))$. Recognising which standard form a problem reduces to is half the battle.

The fourth is the rational ladder. Most Cauchy-type problems follow the same five-step pattern: positivity (or signs), values on $\mathbb{N}$, values on $\mathbb{Z}$, values on $\mathbb{Q}$, then values on $\mathbb{R}$ via continuity or monotonicity. Internalising this template makes such problems mechanical.

End of Note